BSIT MELEC 9: System Need Analysis

Module 1 Study Reviewer & Practice Examination

COURSE: BSIT - MELEC 9 (System Need Analysis)	**REFERENCE:** BSIT_LS_MELEC9_N01.docx (Learning Sheet No. 1)
TOPIC: Introduction to Systems Analysis and Design (SAD)	**COVERAGE:** Prelim Examination
Key Competencies: Define system, analysis, design, SAD; 5 system components; role & skills of a system analyst; importance of SAD.	**Format:** Notes + Practice Test + Answer Key

I. Core Concepts & Foundations

1. What is a System?

A **system** is a collection of interrelated components working together toward a common goal by accepting inputs, processing them, and producing structured outputs.

The Five (5) Components of a System:

Component	Core Function / Definition	School Enrollment System Example
1. Input	Data, materials, or resources entered into the system.	Student personal info, selected subjects, payment slips.
2. Process	Activities and transformations applied to input data.	Validating credentials, calculating tuition, assigning subjects.
3. Output	The resulting information, reports, or services produced.	Certificate of Registration (COR), class schedule, official receipt.
4. Feedback	Evaluation data used to assess performance and make adjustments.	Student verifies enrollment details; system logs error prompts.
5. Control	Rules, permissions, and procedures that govern correct operations.	Only authorized registrar/cashier staff can approve enrollment transactions.

EXAM TIP / CRITICAL CONCEPT: INPUT VS. OUTPUT VS. CONTROL

Exam questions frequently ask you to classify parts of a real-world system. Remember: *Input* is raw data; *Process* is computation/logic; *Output* is finished information; *Feedback* is reaction/metrics; and *Control* is security/authorization rules.

II. Differentiating Analysis from Design

Understanding the fundamental boundary between the **Analysis** phase and the **Design** phase is one of the most critical exam topics:

Criterion	System Analysis (What?)	System Design (How?)
Core Focus	Understanding the problem, existing processes, and business requirements.	Creating the technical blueprint and specification for the solution.
Guiding Question	*'What does the organization and user need the system to do?'*	*'How will the system physically fulfill those requirements?'*
Primary Activities	Interviews, observations, surveys, analyzing legacy documents, workflows.	Designing user interfaces (UI), database schemas (ERD), process models, security.
Primary Deliverables	Problem statement, Software Requirements Specification (SRS), user stories.	Wireframes, mockups, data flow diagrams (DFD), database tables, architecture charts.
Target Audience	Business owners, end-users, domain specialists, management.	Software engineers, database administrators, UI/UX designers, developers.

III. What is System Analysis and Design (SAD)?

System Analysis and Design (SAD) is the systematic process of studying an existing or proposed business system, identifying user requirements, and designing an efficient computer-based solution to solve organizational problems.

The Danger of Skipping SAD vs. Benefits of Applying SAD:

- **Without SAD:** Programmers jump straight into coding without consulting stakeholders. Results: missing vital features, wrong reports, poor user adoption, massive budget overruns, and severe software bugs.

• **With SAD:** Thorough requirements elicitation occurs first. Results: clear roadmap, accurate data modeling, alignment with actual user workflows, robust security, lower maintenance costs, and high user satisfaction.

IV. The Role, Responsibilities, and Skills of a System Analyst

A **System Analyst** is an IT professional who analyzes business problems, gathers stakeholder requirements, and designs information systems that improve business operations. The analyst serves as the vital **bridge / translator** between non-technical business users and technical software developers.

Seven (7) Key Responsibilities of a System Analyst:

1. **Understand Business Problems:** Study legacy workflows to pinpoint bottlenecks, paper-trail delays, or duplicate entries.
2. **Gather User Requirements:** Consult end-users via interviews, questionnaires, active/passive observation, and document analysis.
3. **Analyze Existing Systems:** Evaluate current processes to determine strengths, weaknesses, manual overhead, and compliance risks.
4. **Design the New System:** Construct detailed models including Data Flow Diagrams (DFDs), Entity-Relationship Diagrams (ERDs), and UI mockups.
5. **Communicate with Stakeholders:** Facilitate alignment between clients, department managers, and technical programming teams.
6. **Assist During Development:** Clarify ambiguous requirements, review code implementations against specs, and answer programmer queries.
7. **Participate in System Testing:** Validate that the delivered software functions exactly according to documented acceptance criteria.

Five (5) Essential Skill Categories:

Skill Domain	Key Competencies & Practical Application
1. Technical Skills	Database design (SQL, normalization), system modeling (DFD, ERD, UML), understanding architecture and programming.
2. Analytical Skills	Critical thinking, root-cause analysis, complex problem-solving, data interpretation, logical reasoning.
3. Communication Skills	Verbal clarity, technical writing, active listening, delivering persuasive presentations to executive stakeholders.
4. Interpersonal Skills	Negotiation (resolving conflicting user desires), teamwork, empathy, mediation, stakeholder management.
5. Business Skills	Deep knowledge of organizational operations, accounting/finance basics, cost-benefit analysis, project scheduling.

V. Ten (10) Strategic Benefits of SAD

1. **Identifies True User Needs:** Discovers underlying root causes rather than superficial requests.
2. **Solves Organizational Inefficiencies:** Replaces slow manual workflows with streamlined automated systems.
3. **Reduces Development Errors:** Finding logic/specification defects during analysis costs 10x–50x less than fixing deployed software.
4. **Saves Financial & Human Resources:** Minimizes costly rework, redesign cycles, and wasted developer hours.
5. **Elevates System Quality:** Delivers software that is reliable, scalable, maintainable, and robust.
6. **Boosts User Acceptance:** Users eagerly adopt tools designed specifically around their daily habits and needs.
7. **Empowers Decision-Making:** Provides managers with timely, accurate reporting and executive dashboards.
8. **Enforces Robust Security:** Implements role-based access control (RBAC), data validation, audit logs, and encryption.
9. **Enables Scalability:** Modular architectural design allows seamless future integrations (e.g. mobile apps, payment gateways).
10. **Multiplies Productivity:** Eliminates redundant data entry so employees can focus on high-value business objectives.

MODULE 1: SAMPLE EXAMINATION

Test your mastery of Module 1. Complete all parts before checking the Answer Key.

PART I: MULTIPLE CHOICE (10 Items)

Select the best answer for each question.

1. Which component of an Information System provides the rules, procedures, and authorization safeguards that ensure data integrity?

A. Feedback
B. Process
C. Control
D. Output

2. In a School Enrollment System, what component does a generated Certificate of Registration (COR) represent?

A. Input
B. Output
C. Process
D. Control

3. A developer begins writing source code for a hospital portal immediately without speaking to doctors or nurses. What essential phase was bypassed?

A. System Analysis
B. System Deployment
C. Unit Testing
D. Code Compilation

4. Which question best captures the fundamental objective of the System Design phase?

A. 'Why does the organization exist?'
B. 'What does the user need the system to accomplish?'
C. 'How will the computer-based system physically meet user requirements?'
D. 'How much profit will the business make next year?'

5. Which of the following is considered the primary metaphorical role of a System Analyst in an IT project?

A. A hardware technician who configures cables
B. A bridge / translator between business users and programmers
C. A financial auditor who approves company taxes
D. A database clerk who enters daily student records

6. When an analyst interviews a registrar and discovers that manual paper grading causes student queues, which analyst responsibility is being fulfilled?

A. Designing user interfaces
B. Understanding business problems and gathering requirements
C. Writing back-end database stored procedures
D. Performing automated integration testing

7. Which skill category enables an analyst to resolve disagreements between two department heads who demand conflicting system features?

A. Technical Skills
B. Interpersonal Skills
C. Programming Skills
D. Hardware Architecture Skills

8. Data Flow Diagrams (DFDs) and Entity-Relationship Diagrams (ERDs) are primarily constructed during which phase?

A. System Analysis and Design
B. Hardware Procurement
C. End-User Training
D. Decommissioning

9. Why is detecting and correcting errors during the Analysis phase drastically better than during the Maintenance phase?

A. Fixing errors during analysis costs a fraction of the time and money compared to post-deployment rework.
B. Developers do not get paid during the analysis phase.
C. Users are never affected by bugs in deployed software.
D. Analysis is handled exclusively by automated AI tools.

10. Incorporating Role-Based Access Control (RBAC) so that only cashiers can accept tuition payments directly supports which system component?

A. Control
B. Input
C. Feedback
D. Output

PART II: IDENTIFICATION / TERMINOLOGY (5 Items)

Provide the correct term or concept described in each statement.

11. The component of a system consisting of information or signals used to make adjustments and evaluate system performance: ____________________

12. The specific IT professional responsible for studying business problems, gathering user requirements, and designing solutions: ____________________

13. The developmental phase that serves as the technical 'blueprint' detailing user interfaces, database tables, and system architecture: ____________________

14. The term for data, resources, or raw facts entered into an information system: ____________________

15. The core framework / acronym that encompasses understanding organizational problems and architecting effective computer-based solutions: ____________________

PART III: TRUE OR FALSE (5 Items)

Write TRUE if the statement is correct; write FALSE if the statement is incorrect.

16. System Analysis focuses primarily on 'HOW' the software will be technically coded, while System Design focuses on 'WHAT' the business needs.

17. System Analysts require strong business acumen because software must strictly align with organizational goals and commercial workflows.

18. Skipping System Analysis and Design usually accelerates total project completion and drastically reduces long-term software maintenance costs.

19. Control procedures in a system ensure that only authorized users can perform sensitive transactions such as updating grades or issuing refunds.

20. An effective System Analyst only needs programming knowledge; interpersonal and communication skills are completely optional.

PART IV: SCENARIO APPLICATION & ESSAY (2 Items)

Read each scenario carefully and answer comprehensively based on your learning sheet.

Question 21: A programmer is hired to build a School Enrollment System. Eager to show progress, he immediately begins writing code in Visual Studio Code without speaking with the school registrar, cashiers, teachers, or students. Explain at least **three (3) major problems** that are likely to occur, and describe how applying **System Analysis and Design (SAD)** would prevent each problem.

Question 22: Identify and describe the **five (5) components of a system** in the context of an *Online Food Delivery Application* (such as GrabFood or Foodpanda). For each component, provide one concrete real-world example from the app.

MODULE 1: ANSWER KEY & DETAILED RATIONALES

Question 1. C (Control)
Rationale: Control encompasses rules, authorization safeguards, and access privileges ensuring the system operates securely and validly.

Question 2. B (Output)
Rationale: The Certificate of Registration (COR) is generated information produced by the system for the student and faculty.

Question 3. A (System Analysis)
Rationale: The developer bypassed analyzing the existing problem and eliciting user requirements directly from hospital stakeholders.

Question 4. C ('How will the computer-based system physically meet user requirements?')
Rationale: Analysis defines 'WHAT' must be solved; Design defines 'HOW' the technology will solve it.

Question 5. B (A bridge / translator between business users and programmers)
Rationale: Analysts translate ambiguous operational needs into technical specifications that coders can build.

Question 6. B (Understanding business problems and gathering requirements)
Rationale: Interviewing frontline staff to discover pain points is a core requirements gathering responsibility.

Question 7. B (Interpersonal Skills)
Rationale: Negotiation, diplomacy, and conflict resolution belong to the interpersonal skill domain.

Question 8. A (System Analysis and Design)
Rationale: DFDs and ERDs are standard analytical and design modeling tools used to map data flow and data structures.

Question 9. A (Fixing errors during analysis costs a fraction of the time and money)
Rationale: Studies in software engineering show fixing defects post-release is 10 to 50 times more expensive due to redesign, re-testing, and redeployment.

Question 10. A (Control)
Rationale: Role-based permissions (RBAC) restrict unauthorized actions, functioning as a system control mechanism.

PART II: IDENTIFICATION ANSWERS

11. Feedback — Information returning from the environment or user to adjust or confirm performance.

12. System Analyst — The IT specialist who acts as liaison and solution designer.

13. System Design — The architectural and physical blueprint phase.

14. Input — Raw data and resources injected into the system.

15. System Analysis and Design (SAD) — The overall methodology of studying problems and designing computer solutions.

PART III: TRUE OR FALSE ANSWERS

16. FALSE — Analysis focuses on 'WHAT' the system must do; Design focuses on 'HOW' it will be built.

17. TRUE — Analysts must understand business goals, accounting, and processes to create practical software.

18. FALSE — Skipping SAD leads to massive rework, budget overruns, and expensive post-release maintenance.

19. TRUE — Control mechanisms enforce security, auditing, and authorization rules.

20. FALSE — Communication, negotiation, and interpersonal skills are equally or more vital than programming for an analyst.

PART IV: SCENARIO APPLICATION SAMPLE ANSWERS

Model Answer for Question 21:
Problems: 1) *Missing Crucial Features:* The programmer might omit prerequisites validation or tuition scholarship discounts because he never consulted the registrar. 2) *Usability Failure:* The interface may be too confusing or illogical for non-technical office staff, causing delays. 3) *Faulty Reporting:* Reports generated might not match CHED or school accounting formats.
How SAD Solves This: SAD requires conducting stakeholder interviews first (gathering requirements), constructing UI mockups for early user feedback (design), and establishing clear validation rules before any code is typed.

Model Answer for Question 22 (Food Delivery App):
1. **Input:** Customer delivery address, selected food items, and credit card payment information.
2. **Process:** Calculating subtotal, delivery fee, and taxes; finding nearest delivery rider via GPS matching.
3. **Output:** Digital order receipt, real-time map showing rider location, and order notification to the restaurant.
4. **Feedback:** Star rating and written customer review of the food/delivery; SMS confirmation of delivery.
5. **Control:** OTP / two-factor authentication on checkout; merchant verification before listing a restaurant menu.